VPS Application Deployment Guide

1. Generate SSH Deploy Key

Generate a dedicated deploy key for your GitHub repository:

```shell
ssh-keygen -t ed25519 -C "rhamify-app-deploy" -f
~/.ssh/rhamify_app_deploy -N ""
```

Display and copy the public key to add to GitHub (under **Repository Settings** → **Deploy keys**):

```shell
cat ~/.ssh/rhamify_app_deploy.pub
```

2. Configure SSH Host Alias on Server

Edit or create your SSH configuration file:

```shell
nano ~/.ssh/config
```

Add the following host entry for your application:

```none
Host github-rhamify-app
  HostName github.com
  User git
  IdentityFile ~/.ssh/rhamify_app_deploy
  IdentitiesOnly yes
```

Set secure permissions on the SSH configuration file:

```shell
chmod 600 ~/.ssh/config
```

3. Clone the Repository

Test the SSH connection and clone your repository:

```shell
ssh -T git@github-rhamify-app
git clone git@github-rhamify-app:YOUR_USER/your-repo.git
~/rhamify_app
cd ~/rhamify_app
git config core.fileMode false
```

4. Configure Environment and Build Dependencies

Set up the environment variables, backend, database, and asset compilation:

```shell
cp .env.example .env
nano .env
composer install --no-dev --optimize-autoloader
php artisan key:generate
php artisan migrate --force
php artisan storage:link
npm ci && npm run build
```

5. Set File and Web Server Permissions

Grant access permissions for the Nginx user and set required permissions for Laravel storage and cache:

```shell
setfacl -m u:ploi:x ~/rhamify_app ~/rhamify_app/storage
~/rhamify_app/storage/app
setfacl -R -m u:ploi:rX ~/rhamify_app/public
~/rhamify_app/storage/app/public
setfacl -R -d -m u:ploi:rX ~/rhamify_app/public
~/rhamify_app/storage/app/public
chmod -R 775 ~/rhamify_app/storage ~/rhamify_app/bootstrap/cache
```

6. Configure PHP-FPM

Create and edit the PHP-FPM pool configuration for your application:

Shell

```
sudo nano /etc/php/8.4/fpm/pool.d/rhamify_app.conf
```

Test and reload the PHP-FPM service:

Shell

```
sudo php-fpm8.4 -t
sudo systemctl reload php8.4-fpm
```

7. Configure Nginx and SSL Certificate

Create the Nginx site configuration at `/etc/nginx/sites-available/rhamify_app` (Stage 1 - HTTP port 80 only):

None

```
server {
    listen 80;
    listen [::]:80;
    server_name new.rhamify.com;
    root /home/rhamifytechnology/rhamify_app/public;
    index index.php;
    client_max_body_size 20M;
```

None

```
    location / {
        try_files $uri $uri/ /index.php?$query_string;
    }
```

None

```
    location ~ \.php$ {
        fastcgi_pass unix:/run/php/rhamify_app-fpm.sock;
        fastcgi_param SCRIPT_FILENAME
$realpath_root$fastcgi_script_name;
        include fastcgi_params;
        fastcgi_buffer_size 32k;
        fastcgi_buffers 16 16k;
```

```
        fastcgi_busy_buffers_size 32k;
    }
```

```
None
    location ~ /\.(?!well-known).* {
        deny all;
    }
}
```

Obtain an SSL certificate using Certbot:

```
Shell
sudo certbot certonly --webroot -w
/home/rhamifytechnology/rhamify_app/public -d new.rhamify.com
--deploy-hook "systemctl reload nginx"
```

Always verify Nginx configuration syntax before reloading:

```
Shell
sudo nginx -t
```

8. Configure Queue Worker & Reverb Process Supervisor

Create a Supervisor configuration file at `/etc/supervisor/conf.d/rhamify_app.conf`:

```
None
[program:rhamify_app-queue]
command=php /home/rhamifytechnology/rhamify_app/artisan
queue:work --sleep=3 --tries=3 --max-time=3600
user=rhamifytechnology
autostart=true
autorestart=true
stopasgroup=true
killasgroup=true
```

```
None
[program:rhamify_app-reverb]
command=php /home/rhamifytechnology/rhamify_app/artisan
reverb:start --port=8082
user=rhamifytechnology
autostart=true
autorestart=true
```

Update Supervisor to manage the new processes:

```
Shell
sudo supervisorctl reread && sudo supervisorctl update
```

9. Set Up Auto-Deployment Script

Create the auto-deployment script at `~/rhamify_app/deploy.sh` and make it executable:

```
Shell
#!/bin/bash
set -e
cd /home/rhamifytechnology/rhamify_app
```

```
Shell
php artisan down || true
trap 'php artisan up' EXIT
```

```
Shell
git pull origin main
composer install --no-dev --optimize-autoloader --no-interaction
npm ci && npm run build
php artisan migrate --force
php artisan optimize
php artisan queue:restart
php artisan reverb:restart || true
```

Make the script executable and exclude it from version tracking:

```shell
chmod +x ~/rhamify_app/deploy.sh
echo "deploy.sh" >> .git/info/exclude
```

10. Configure GitHub Actions Workflow

Add a GitHub Actions workflow file at `.github/workflows/deploy.yml` in your repository to execute `/home/rhamifytechnology/rhamify_app/deploy.sh` via SSH on pushes to `main`.
Configure the following Repository Secrets under **Settings** → **Secrets and variables** → **Actions** → **Secrets**:

- `SSH_HOST`: VPS IP address or hostname.
- `SSH_USER`: SSH user account (e.g., `rhamifytechnology`).

`SSH_KEY`: Private SSH key with access to the server.